Task 2: Zomato Dataset Analysis Report

1. Executive Summary

This report presents an exploratory data analysis of a Zomato restaurant dataset. The objective of this task was to analyze restaurant and review-related data to identify insights about ratings, cuisines, restaurant locations, price patterns, online ordering, table booking, and factors that may influence customer ratings.

The dataset contained restaurant information such as name, address, location, restaurant type, cuisines, ratings, votes, online ordering availability, table booking availability, approximate cost for two people, and listing category. The data required cleaning because some columns contained missing values, text-based ratings, cost values with commas, and a few misaligned or corrupted rows.

After cleaning and preprocessing, the analysis focused on rating distribution, popular restaurant locations, cuisine popularity, cuisine ratings, location-based rating patterns, price vs rating relationship, online order impact, table booking impact, and correlation among numeric variables.

The average restaurant rating in the cleaned dataset was **3.72**. The most common restaurant location was **BTM**, while the most popular cuisine was **North Indian**. Among common cuisines, **Malaysian** showed the highest average rating. **Lavelle Road** showed the highest average rating among active restaurant locations. Restaurants with **online ordering** and **table booking** availability showed higher average ratings, suggesting that convenience features may improve customer experience and restaurant performance.

Based on these findings, practical recommendations were prepared for an Alfido Tech style restaurant or food discovery platform.

2. Objective

The objective of this analysis is to study Zomato restaurant data and extract useful insights related to:

- Restaurant ratings
- Popular cuisines
- Location preferences
- Restaurant hotspots
- Price vs rating relationship
- Online ordering impact
- Table booking impact
- Factors affecting customer ratings

- Business recommendations for an Alfido Tech style platform

The goal is to convert raw restaurant data into meaningful insights that can support better restaurant discovery, content planning, partnerships, and user-focused recommendations.

3. Dataset Overview

The dataset contained restaurant-related information from Zomato. It included details such as restaurant names, addresses, locations, ratings, votes, cuisines, restaurant type, online ordering availability, table booking availability, and approximate cost for two people.

Main Columns

Column	Description
address	Restaurant address
name	Restaurant name
online_order	Whether online ordering is available
book_table	Whether table booking is available
rate	Restaurant rating
votes	Number of votes received
phone	Restaurant contact information
location	Restaurant location
rest_type	Type of restaurant
dish_liked	Popular dishes liked by customers
cuisines	Cuisines served
approx_cost(for two people)	Approximate cost for two people
listed_in(type)	Listing category or restaurant service type

The dataset was useful for understanding how restaurant features, location, cuisine, and customer convenience options relate to ratings and popularity.

4. Tools Used

The following tools and technologies were used for this task:

- Python
- Pandas
- NumPy
- Matplotlib

- Seaborn
- WordCloud
- Google Colab
- GitHub

5. Data Cleaning and Preprocessing

The raw dataset required cleaning before analysis. The following preprocessing steps were performed:

Column Name Cleaning

Column names were standardized by converting them to lowercase and replacing spaces with underscores. This made the dataset easier to work with in Python.

Rating Column Cleaning

The rating column contained values such as ratings in text format, missing values, and non-rating entries. These values were cleaned by:

- Removing /5 from rating values
- Replacing invalid values such as NEW and -
- Converting ratings into numeric format
- Filling missing ratings using the median rating

Cost Column Cleaning

The approximate cost column contained comma-separated values. It was cleaned by:

- Removing commas
- Converting values into numeric format
- Filling missing cost values using the median cost

Missing Value Handling

Missing values in important text columns such as location, restaurant type, cuisines, dishes liked, listing type, online order, and table booking were filled with Unknown.

Duplicate Removal

Duplicate rows were checked and removed to avoid repeated records affecting the analysis.

Corrupted Row Filtering

Some rows had misaligned values due to long review text appearing in the wrong columns. Rows where important columns such as online order and table booking did not contain valid Yes or No values were filtered out.

This cleaning step was important because invalid rows could affect charts and average rating calculations.

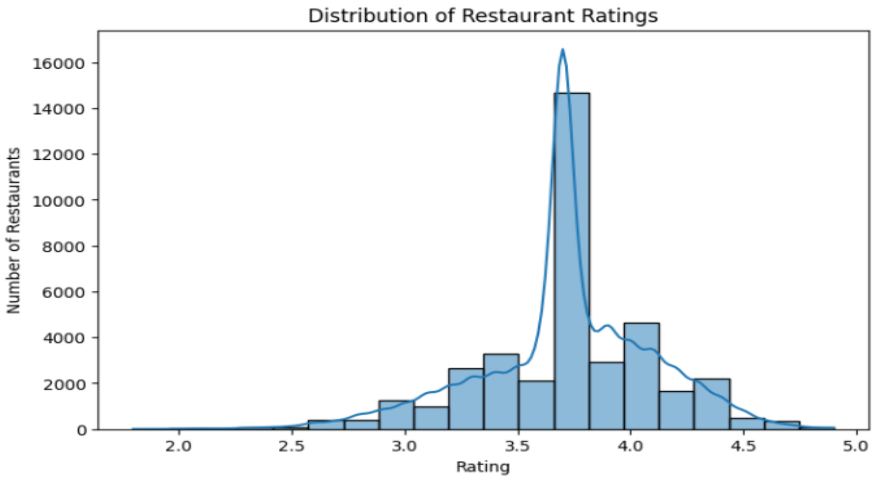
6. Analysis Performed

The following analyses were performed:

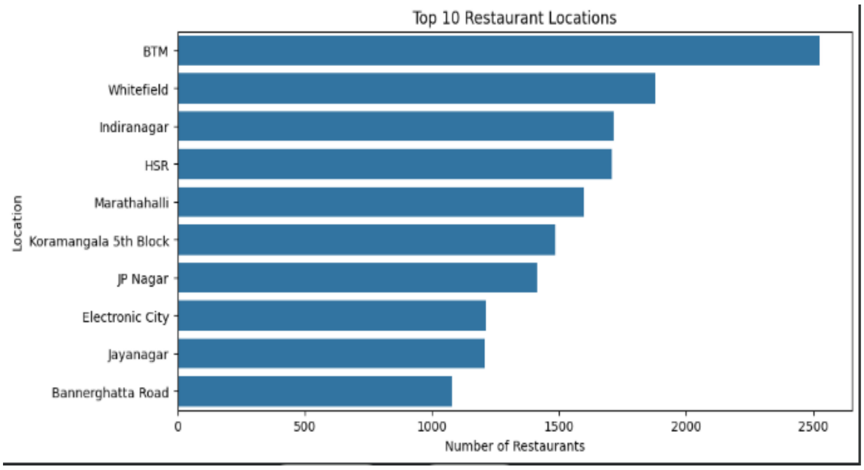
- Rating distribution analysis
- Top restaurant location analysis
- Online ordering vs average rating
- Table booking vs average rating
- Price vs rating relationship
- Popular restaurant type analysis
- Popular cuisine analysis
- Cuisine vs average rating analysis
- Location vs average rating analysis
- Correlation heatmap for numeric features
- Wordcloud for popular cuisines
- Key insight extraction
- Business recommendations for Alfido Tech style platform

7. Key Visualizations

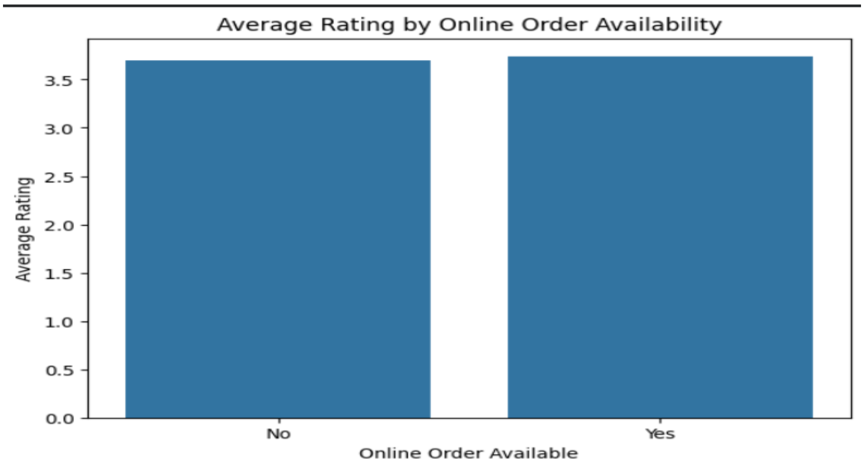
7.1 Rating Distribution



7.2 Top 10 Restaurant Locations

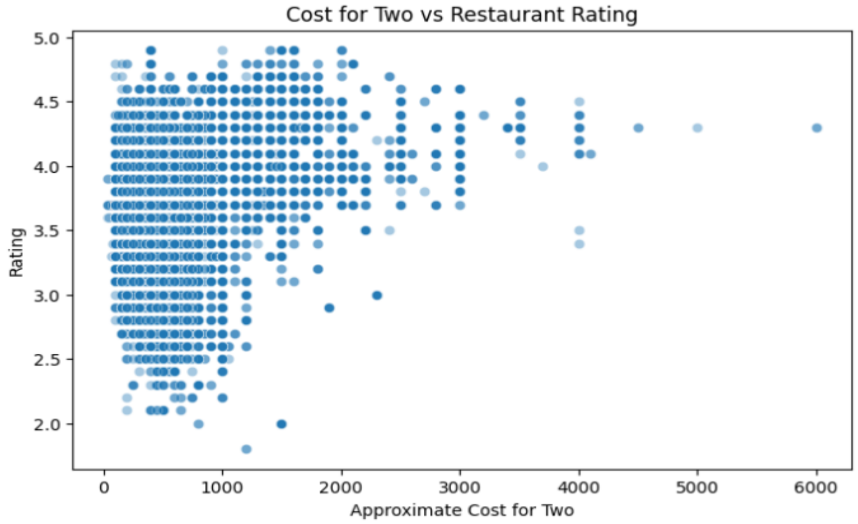


7.3 Online Order vs Average Rating

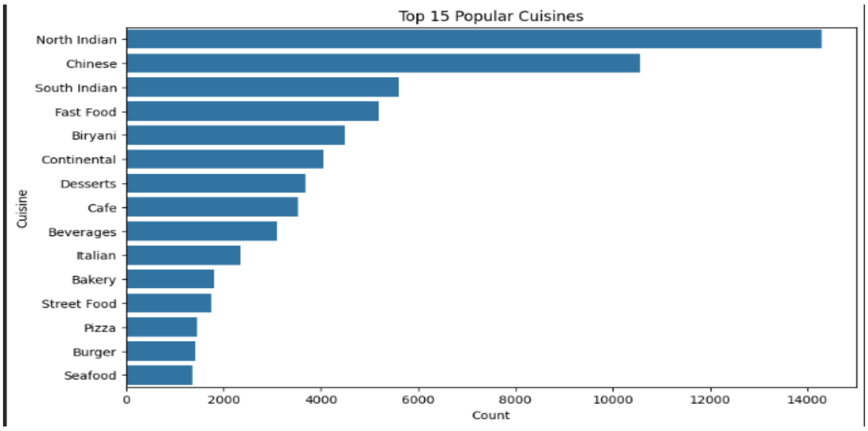


7.4 Table Booking vs Average Rating

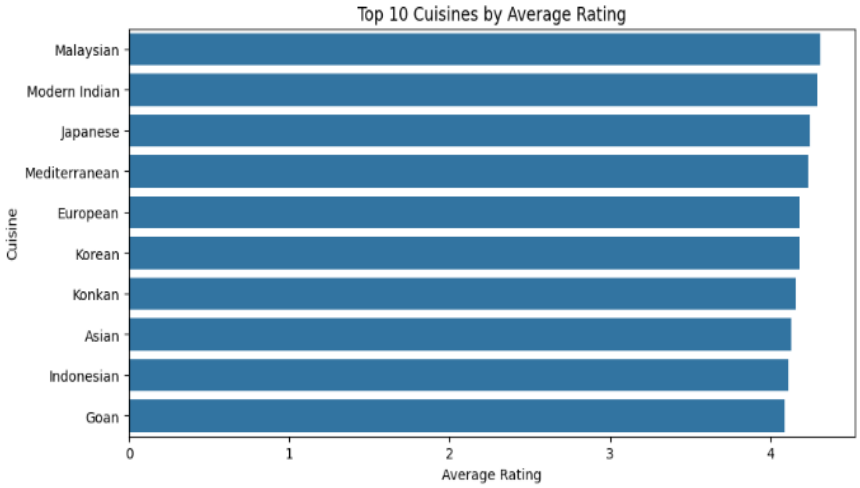
7.5 Cost for Two vs Rating



7.6 Top 15 Popular Cuisines



7.7 Top 10 Cuisines by Average Rating



7.11 Key Insights Output

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KEY INSIGHTS
------------------------------------------------------------
Average restaurant rating: 3.72
Most common restaurant location: BTM
Most popular cuisine: North Indian
Highest-rated cuisine among common cuisines: Malaysian
Highest-rated location among active locations: Lavelle Road
Online order category with higher average rating: Yes
Table booking category with higher average rating: Yes
```

8. Key Findings

Analysis Area	Finding
Average Restaurant Rating	3.72
Most Common Restaurant Location	BTM
Most Popular Cuisine	North Indian
Highest-Rated Cuisine Among Common Cuisines	Malaysian
Highest-Rated Active Location	Lavelle Road
Online Order Category with Higher Rating	Yes
Table Booking Category with Higher Rating	Yes

Interpretation of Findings

The average rating of **3.72** indicates that most restaurants in the dataset are rated around the moderate to good range.

BTM appeared as the most common restaurant location, which suggests that it is a restaurant-dense area and may be a strong location for food discovery or partnership opportunities.

North Indian was the most popular cuisine, showing strong customer demand and wide availability.

Among common cuisines, **Malaysian** had the highest average rating. This may indicate that niche or less common cuisines can perform strongly when quality and customer experience are good.

Lavelle Road showed the highest average rating among active locations, suggesting it may be a premium or high-performing restaurant area.

Restaurants with **online ordering** and **table booking** availability showed higher average ratings, which suggests that convenience-based features may improve customer satisfaction and overall restaurant performance.

9. Recommendations for Alfido Tech Style Platform

Based on the analysis, the following recommendations can be made:

1. Focus partnerships in high-density restaurant locations

Locations like **BTM**, where many restaurants are present, can be useful for partnerships, restaurant onboarding, campaigns, and food discovery features.

2. Promote popular cuisines through curated content

Since **North Indian** is the most popular cuisine, the platform can create curated lists such as:

- Best North Indian restaurants
- Budget North Indian meals
- Top-rated North Indian places
- Student-friendly North Indian food spots

3. Highlight high-rated niche cuisines

The analysis showed that **Malaysian** cuisine had a high average rating among common cuisines. Alfido Tech can promote niche cuisines through special discovery sections such as:

- Hidden gems
- Try something new
- High-rated unique cuisines

4. Use location-based recommendations

Since location strongly affects restaurant discovery, the platform can provide location-based suggestions such as:

- Best restaurants near you
- Top-rated places in Lavelle Road
- Popular food spots in BTM
- Budget restaurants by area

5. Encourage restaurants to support online ordering and table booking

Restaurants with online ordering and table booking availability showed higher average ratings. These features can be promoted as filters or badges on the platform to improve user convenience.

Example platform filters:

- Online order available

- Table booking available
- High-rated restaurants
- Budget-friendly restaurants
- Popular cuisines nearby

10. Business Interpretation

For a food or restaurant discovery platform, the analysis shows that restaurant performance is influenced by multiple factors, including location, cuisine, convenience features, ratings, and price.

A platform like Alfido Tech can use these insights to improve user experience by recommending restaurants based on:

- User location
- Preferred cuisine
- Rating level
- Budget
- Online ordering availability
- Table booking availability

The platform can also support restaurants by identifying high-demand cuisines and high-performing areas. This can help in marketing campaigns, restaurant partnerships, and personalized recommendations.

The analysis also suggests that user convenience features such as online ordering and table booking may increase customer satisfaction, so they should be highlighted clearly in the platform interface.

11. Limitation

The dataset required significant cleaning because some rows had missing values, invalid rating formats, and misaligned text values. These issues may slightly affect analysis accuracy.

The dataset also does not include full customer-level behavior such as user demographics, repeat orders, review sentiment scores, delivery time, or actual transaction data.

Therefore, the insights should be treated as exploratory findings. For production-level decisions, more recent and cleaner restaurant data would be required.

12. LINKS

GitHub Task Folder:

https://github.com/saitharun-student-27/InternSpark-Data-Science-Internship/tree/main/Task-2-Zomato-Dataset-Analysis

Notebook Link:
https://colab.research.google.com/drive/17KHKbYqDzH4JltDccHnk1N-YpQWrbIwx?usp=sharing

13. Conclusion

This task helped analyze real restaurant data and understand how ratings, cuisines, locations, prices, and convenience features affect restaurant performance.

Through this task, the dataset was cleaned, missing values were handled, corrupted rows were filtered, and multiple visualizations were created to extract insights. The analysis showed that **BTM** was the most common restaurant location, **North Indian** was the most popular cuisine, **Malaysian** was the highest-rated cuisine among common cuisines, and **Lavelle Road** had the highest average rating among active locations.

The main learning from this task was that data analysis is not only about creating charts. It is about cleaning messy data, identifying useful patterns, interpreting results, and converting them into practical business recommendations.